

NON-COMMUTATIVE DYNAMICS ON MANIFOLDS

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ABSTRACT. Conventional approaches to complex-systems dynamics often represent the temporal evolution of macroscopic observables within commutative state spaces, typically relying on diffusion-based stochastic models on flat Euclidean geometries. Such formulations, while effective for a broad class of systems, are inherently limited in their ability to represent non-commutative dynamics and path dependence, in which the ordering and historical composition of discrete events can critically influence the subsequent trajectory of the system. To overcome this limitation, we introduce a general axiomatic framework for dynamical systems in which sequences of discrete, non-commutative micro-events are encoded as algebraic words and subsequently induce geometric and topological transformations of the underlying state space. In this framework, the emergent manifolds provide a geometric representation of the global structure generated by microscopic event sequences, thereby establishing a systematic correspondence between discrete event dynamics and emergent geometric structure, while providing a topological representation that can distinguish non-commutative path-dependent compositions of event sequences.

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1. INTRODUCTION

Let G be a finitely generated non-abelian group with a finite generating set $S = \{s_1, s_2, \dots, s_n\}$, and let $\Gamma(G, S)$ denote its associated Cayley graph. Many commonly used formulations of complex-system dynamics represent macroscopic evolution using stochastic processes on commutative state spaces, typically relying on diffusion models over flat Euclidean geometries. While highly effective for standard continuous-space phenomena, formulating a direct structural bridge from discrete, non-commutative micro-event compositions to macroscopic manifold topologies remains a key challenge. To bridge this gap, we propose a topological framework that encodes temporal sequences of non-abelian micro-events as algebraic words, which sequentially induce mapping class transformations on a reference surface Σ . By constructing a discrete dynamical system whose trajectory induces a time-indexed sequence of mapping tori $\{M_t\}_{t \geq 1}$, we establish a formal correspondence between non-commutative discrete event compositions and dynamic topological state spaces.

Formal picture. The foundational architecture rests on treating the sequence of discrete events as a word in the free group $F(S)$, which is subsequently projected to G , which maps to a topological transformation on a reference surface Σ . By constructing mapping tori and analyzing stochastic processes on the resulting spaces, we replace static background geometries with emergent manifolds whose topological invariants capture the non-commutative path-dependent composition of event sequences.

Remark on Information Flow. The algebraic-topological pipeline

$$F(S) \rightarrow G \rightarrow \text{MCG}(\Sigma) \rightarrow \mathcal{M}(\Sigma)$$

involves successive quotient projections (governed by the structural relations R) and representation maps, where $\mathcal{M}(\Sigma) = \{\text{homeomorphism classes of mapping tori}\}$. Rather than retaining every microscopic permutation, this multi-stage mapping acts as a natural coarse-graining mechanism: it filters out redundant microscopic information while preserving non-commutative ordering whenever that ordering remains distinguishable under the representation and mapping-torus constructions.

$$(1) \quad G \xrightarrow{\rho} \text{MCG}(\Sigma) \xrightarrow{\psi} \{\text{homeomorphism classes of mapping tori}\}$$

$$g \longmapsto [\phi_g] \longmapsto [M_{\phi_g}]$$

Let

$$\rho : G \longrightarrow \text{MCG}(\Sigma)$$

be a group homomorphism encoding the action of discrete micro-events on a closed, connected, orientable reference surface Σ . For each $g \in G$, let $\phi_g : \Sigma \rightarrow \Sigma$ be a representative homeomorphism of the mapping class

$$\rho(g) = [\phi_g].$$

The corresponding mapping torus is then defined by

$$M_{\phi_g} = \frac{\Sigma \times [0, 1]}{(x, 0) \sim (\phi_g(x), 1)}.$$

Thus, the proposed construction yields the correspondence

$$G \xrightarrow{\rho} \text{MCG}(\Sigma) \xrightarrow{\psi} \{\text{homeomorphism classes of mapping tori}\}, \quad g \mapsto [\phi_g] \mapsto [M_{\phi_g}],$$

where ψ denotes the mapping-torus construction.

Pipeline Architecture. The construction proceeds in stages: raw continuous time series are discretized into symbolic event sequences generated by S , which are governed by the group presentation $G = \langle S \mid R \rangle$. A representation homomorphism ρ then maps these algebraic relations into the mapping class group $\text{MCG}(\Sigma)$, whose elements induce homeomorphisms that build the emergent 3-manifold M_ϕ .

1.1. Main Research Questions. To establish this non-commutative dynamical framework, this paper explicitly addresses the following five central questions:

- Q1 (Stochastic Mapping Class Dynamics)::** How does a stochastic sequence of non-commutative micro-events induce a well-defined stochastic process on the mapping class group $\text{MCG}(\Sigma)$?
- Q2 (Information Loss & Coarse-Graining)::** How do microscopic non-commutative ordering differences vanish or persist through the coarse-graining hierarchy $\sim_G \subseteq \sim_{\text{MCG}} \subseteq \sim_{\text{top}}$?
- Q3 (Complexity Growth & Entropy Transfer)::** How does the algebraic word growth $d_S(e, G_n)$ of random event sequences quantitatively dictate the topological entropy $h_{\text{top}}(\phi_n)$ of the induced mapping classes?
- Q4 (Asymptotic Hyperbolicity)::** Under what conditions does a random walk on G exhibit asymptotically generic pseudo-Anosov behavior with probability 1?
- Q5 (Emergent Topological Phase Transitions)::** Can a parameterized stochastic micro-event distribution generate a genuine geometric phase transition in the resulting 3-manifold state space?

2. PRELIMINARIES AND ALGEBRAIC FRAMEWORK

To establish a rigorous foundation for non-commutative system dynamics, we formalize the algebraic structure governing discrete micro-events.

2.1. Groups and Word Representations. Let G be a finitely generated group defined by a presentation $\langle S \mid R \rangle$, where $S = \{s_1, s_2, \dots, s_n\}$ represents the set of fundamental microscopic events, and R denotes the set of structural relations or equivalence constraints.

Definition 2.1 (Cayley Graph and Cayley Complex). Let $G = \langle S \mid R \rangle$ be a finitely presented group. The Cayley graph $\Gamma(G, S)$ is the directed, S -labeled graph whose vertices are the elements of G , with an edge

$$g \xrightarrow{s} gs$$

for each $g \in G$ and $s \in S$.

The Cayley 2-complex $X(G, S, R)$ is obtained from $\Gamma(G, S)$ by attaching a 2-cell along every closed path whose edge-label is a relator $r \in R$ (and its translates in G).

The combinatorial path length on the Cayley graph induces the word metric

$$d_S(g, h) = \min\{|w| : w \in (S \cup S^{-1})^*, g^{-1}h = \bar{w}\},$$

where $|w|$ denotes the length of the word w over the symmetric generating set $S \cup S^{-1}$, providing a natural measure of the combinatorial length of an event sequence.

2.2. Mapping Class Groups and Emergent Manifolds.

Definition 2.2 (Mapping Class Group). Let Σ be a closed, connected, orientable surface serving as the reference space on which the micro-events act. The **mapping class group** $\text{MCG}(\Sigma)$ is defined as the group of isotopy classes of orientation-preserving homeomorphisms of Σ :

$$\text{MCG}(\Sigma) = \frac{\text{Homeo}^+(\Sigma)}{\text{Homeo}_0^+(\Sigma)},$$

where $\text{Homeo}_0^+(\Sigma)$ denotes the subgroup of orientation-preserving homeomorphisms isotopic to the identity.

Definition 2.3 (Mapping Torus). For each $g \in G$, choose a representative homeomorphism $\phi_g : \Sigma \rightarrow \Sigma$ of the mapping class $\rho(g)$. The mapping torus associated with ϕ_g is defined by

$$M_{\phi_g} = \frac{\Sigma \times [0, 1]}{(x, 0) \sim (\phi_g(x), 1)}.$$

The mapping-torus construction induces a well-defined set mapping

$$\psi : \text{MCG}(\Sigma) \longrightarrow \{\text{homeomorphism classes of mapping tori}\}$$

defined by $\psi([\phi]) = [M_\phi]$. The well-definedness of ψ follows from the isotopy invariance proved in the subsequent proposition.

Proposition 2.4 (Topological Nature of Mapping Tori). *Let Σ be a closed, connected, orientable surface, and let $\phi : \Sigma \rightarrow \Sigma$ be an orientation-preserving homeomorphism. Then the resulting mapping torus M_ϕ is a closed, connected, orientable 3-manifold.*

Proof. The surface Σ is compact and connected, and the interval $[0, 1]$ is compact and connected; hence their product $\Sigma \times [0, 1]$ is a compact, connected 3-manifold with boundary $\Sigma \times \{0\} \cup \Sigma \times \{1\}$. The mapping torus M_ϕ is formed by identifying the boundary component $\Sigma \times \{0\}$ with $\Sigma \times \{1\}$ via the homeomorphism ϕ . Because ϕ is a homeomorphism, standard collar-neighborhood gluing ensures that points on the boundary have open neighborhoods homeomorphic to open balls in $\mathbb{R}^3$ rather than half-spaces, thereby eliminating the boundary. Thus, the quotient space is a compact manifold without boundary (a closed 3-manifold) and remains connected. Furthermore, the orientability of M_ϕ is guaranteed by the orientability of Σ combined with the assumption that ϕ is orientation-preserving. $\square$

Remark. It is important to note that ψ is *not* a group homomorphism, as the set of homeomorphism classes of mapping tori does not naturally inherit a group structure compatible with the composition in $\text{MCG}(\Sigma)$. Furthermore, while $[\phi] = [\phi']$ implies $M_\phi \cong M_{\phi'}$, the converse is generally false: distinct mapping classes can occasionally yield homeomorphic mapping tori. Thus, the mapping-torus map ψ inherently entails a degree of topological coarse-graining.

2.3. Well-Definedness of the Mapping Torus Construction. To ensure that the mapping torus M_{ϕ_g} depends solely on the mapping class $[\phi_g] \in \text{MCG}(\Sigma)$ rather than the specific choice of representative homeomorphism ϕ_g , we verify its invariance under isotopy.

Proposition 2.5 (Invariance Under Isotopy). *Let $\phi, \phi' : \Sigma \rightarrow \Sigma$ be two orientation-preserving homeomorphisms. If ϕ is isotopic to ϕ' , then their corresponding mapping tori M_ϕ and $M_{\phi'}$ are homeomorphic.*

Proof. Let $\{\Phi_t\}_{t \in [0,1]}$ be an isotopy from ϕ to ϕ' , satisfying $\Phi_0 = \phi$ and $\Phi_1 = \phi'$. We define a family of homeomorphisms on Σ by

$$h_t = \Phi_t \circ \phi^{-1},$$

which yields $h_0 = \text{id}_\Sigma$ and $h_1 = \phi' \circ \phi^{-1}$, satisfying the relation $h_1 \circ \phi = \phi'$.

We then define a global map $H : \Sigma \times [0, 1] \rightarrow \Sigma \times [0, 1]$ by

$$H(x, t) = (h_t(x), t).$$

Since each h_t is a homeomorphism, H is a homeomorphism of $\Sigma \times [0, 1]$. We check its compatibility with the quotient identifications: at $t = 0$,

$$H(x, 0) = (h_0(x), 0) = (x, 0),$$

and at $t = 1$, evaluating at $\phi(x)$ gives

$$H(\phi(x), 1) = (h_1(\phi(x)), 1) = (\phi'(\phi^{-1}(\phi(x))), 1) = (\phi'(x), 1).$$

Thus, the equivalence relation $(x, 0) \sim (\phi(x), 1)$ in M_ϕ is mapped precisely to $(x, 0) \sim (\phi'(x), 1)$ in $M_{\phi'}$. Therefore, H descends to a well-defined homeomorphism between the quotient spaces:

$$\bar{H} : M_\phi = \frac{\Sigma \times [0, 1]}{(x, 0) \sim (\phi(x), 1)} \xrightarrow{\cong} \frac{\Sigma \times [0, 1]}{(x, 0) \sim (\phi'(x), 1)} = M_{\phi'},$$

proving that the mapping torus structure depends exclusively on the mapping class $[\phi]$. $\square$

2.4. Micro-Event Actions and Group Representation. Let $S = \{s_1, \dots, s_n\}$ be the generating set of G , where each generator represents an elementary micro-event. Let $F(S)$ denote the free group generated by S .

To rigorously disentangle pointwise spatial actions on Σ from isotopy classes in $\text{MCG}(\Sigma)$, we first assign to each generator $s_i \in S$ a specific orientation-preserving homeomorphism $\phi_i \in \text{Homeo}^+(\Sigma)$. By the universal property of free groups, this assignment extends uniquely to a group homomorphism:

$$\tilde{\Phi} : F(S) \longrightarrow \text{Homeo}^+(\Sigma), \quad \text{where } \tilde{\Phi}(s_i) = \phi_i.$$

Composing $\tilde{\Phi}$ with the natural quotient projection $[\cdot] : \text{Homeo}^+(\Sigma) \rightarrow \text{MCG}(\Sigma)$ yields the mapping class representation of algebraic words:

$$\Phi := [\cdot] \circ \tilde{\Phi} : F(S) \longrightarrow \text{MCG}(\Sigma), \quad \Phi(s_i) = [\phi_i].$$

Theorem 2.6 (Induced Mapping Class Representation). *Let $G = \langle S \mid R \rangle$ be a finitely presented group. Suppose that the homomorphism $\Phi : F(S) \rightarrow \text{MCG}(\Sigma)$ satisfies $\Phi(r) = 1$ for every relator $r \in R$. Then Φ factors uniquely through $G = F(S)/\langle\langle R \rangle\rangle$, inducing a well-defined group homomorphism:*

$$\rho : G \longrightarrow \text{MCG}(\Sigma).$$

Proof. Since every relator $r \in R$ satisfies $\Phi(r) = 1 \in \text{MCG}(\Sigma)$, the normal closure $\langle\langle R \rangle\rangle \subseteq \ker \Phi$. By the universal property of quotient groups, there exists a unique group homomorphism $\rho : G \rightarrow \text{MCG}(\Sigma)$ such that $\Phi = \rho \circ \pi$, where $\pi : F(S) \rightarrow G$ is the canonical quotient map. $\square$

2.5. Symplectic Homology Representation and Matrix Order Consistency. To project the non-commutative surface dynamics onto linearized algebraic invariants—specifically the first integral homology $H_1(\Sigma; \mathbb{Z}) \cong \mathbb{Z}^{2g}$ —we formalize the matrix representation induced by micro-event words.

Definition 2.7 (Symplectic Homology Representation). Let Σ be a closed orientable surface of genus $g \geq 1$. The natural action of $\text{MCG}(\Sigma)$ on $H_1(\Sigma; \mathbb{Z}) \cong \mathbb{Z}^{2g}$ equipped with the standard intersection form preserves the symplectic pairing, inducing the canonical symplectic representation:

$$\sigma : \text{MCG}(\Sigma) \longrightarrow \text{Sp}(2g, \mathbb{Z}).$$

Definition 2.8 (Homomorphic Matrix Assignment). To guarantee full algebraic compatibility with the representation homomorphism rule $\rho(w_1 w_2) = \rho(w_1) \rho(w_2)$, the word-to-matrix mapping $M : F(S) \rightarrow \text{Sp}(2g, \mathbb{Z})$ is defined directly without matrix transposition as:

$$M(w) := \sigma(\Phi(w)).$$

This definition ensures that M is a genuine group homomorphism satisfying:

$$M(w_1 w_2) = M(w_1) M(w_2) \quad \forall w_1, w_2 \in F(S).$$

Proposition 2.9 (Chronological Matrix Product for Stochastic Trajectories). *Under the homomorphic matrix assignment defined above, for any cumulative word trajectory $W_n = X_1 X_2 \cdots X_n$ with $X_i \in S^{\pm 1}$, the induced symplectic matrix $M(W_n) \in \text{Sp}(2g, \mathbb{Z})$ decomposes into an ordered product of fundamental event matrices:*

$$M(W_n) = M(X_1) M(X_2) \cdots M(X_n).$$

Consequently, the first homology group of the mapping torus M_n , given by

$$H_1(M_n; \mathbb{Z}) \cong \mathbb{Z} \oplus \text{coker}(M(W_n) - I),$$

is algebraically well-defined and respects the temporal order of microscopic events without matrix order inversion.

2.6. Non-Commutative Event Composition and Hierarchical Distinguishability. The non-commutative and path-dependent structure of our framework operates across three distinct hierarchical strata, which must be systematically distinguished:

- (1) **Algebraic Level:** Two event sequences corresponding to words $w_1, w_2 \in G$ may be non-commutative, i.e., $w_1 w_2 \neq w_2 w_1$ in the group G .
- (2) **Representation Level:** The homomorphism $\rho : G \rightarrow \text{MCG}(\Sigma)$ maps these words to mapping classes $\rho(w_1)$ and $\rho(w_2)$. Due to the potential non-injectivity of ρ , distinct group elements can occasionally project to the same mapping class.
- (3) **Topological Level:** The mapping-torus map yields homeomorphism classes

$$\psi(\rho(w_1)), \quad \psi(\rho(w_2)),$$

whose topological distinction determines macroscopic distinguishability.

To ensure that microscopic event orderings translate into distinct macroscopic geometries, we examine the propagation of non-commutativity through these tiers. Let $s_i, s_j \in S$ such that $s_i s_j \neq s_j s_i$ in G . Under the representation ρ , these generate the mapping classes

$$\rho(s_i s_j) = [\phi_i][\phi_j] \quad \text{and} \quad \rho(s_j s_i) = [\phi_j][\phi_i].$$

Path dependence is preserved at the macroscopic level whenever the composition satisfies:

$$\rho(s_i s_j) \neq \rho(s_j s_i) \quad \text{and} \quad \psi(\rho(s_i s_j)) \neq \psi(\rho(s_j s_i)).$$

This multi-tier hierarchy clarifies that while algebraic event orderings provide the microscopic foundation, macroscopic topological distinctness is secured precisely when the representation and mapping-torus maps avoid accidental homeomorphic collapse.

3. DISCRETE MICRO-EVENT DYNAMICS AND EMERGENT TRAJECTORIES

To transition from a static geometric correspondence to a formal dynamical system, we introduce discrete-time event trajectories and their induced sequence of state transformations.

Definition 3.1 (Event Stream and Cumulative Group Trajectory). Let $T = \mathbb{N}_0 = \{0, 1, 2, \dots\}$ be the discrete time index, and let $S = \{s_1, \dots, s_n\}$ be the generating set of microscopic events.

- (1) An **event stream** (or micro-event trajectory) is a sequence

$$\mathbf{s} = (s_{\alpha_1}, s_{\alpha_2}, s_{\alpha_3}, \dots) \in S^{\mathbb{N}}.$$

- (2) For each time step $t \in \mathbb{N}$, the **cumulative event word** $w_t \in F(S)$ is defined as

$$w_0 = e, \quad w_t = s_{\alpha_1} s_{\alpha_2} \cdots s_{\alpha_t}.$$

- (3) The corresponding **cumulative group state** at time t is given by the canonical projection $g_t = [w_t] \in G = F(S)/\langle\langle R \rangle\rangle$.

Definition 3.2 (Discrete Micro-Event Dynamical System). A **discrete micro-event dynamical system** is a tuple $(\Sigma, G, \rho, \tilde{\Phi}, \mathbf{s})$, where:

- (1) Σ is a closed, connected, orientable surface representing the reference state space;
- (2) $G = \langle S \mid R \rangle$ is the finitely presented event group;
- (3) $\tilde{\Phi} : F(S) \rightarrow \text{Homeo}^+(\Sigma)$ is the actual homeomorphism lift assignment satisfying $\tilde{\Phi}(w_1 w_2) = \tilde{\Phi}(w_1) \circ \tilde{\Phi}(w_2)$;
- (4) $\rho : G \rightarrow \text{MCG}(\Sigma)$ is the induced mapping class representation satisfying $\rho([w]) = [\tilde{\Phi}(w)]$;
- (5) $\mathbf{s} = (s_{\alpha_1}, s_{\alpha_2}, \dots) \in S^{\mathbb{N}}$ is an input micro-event stream.

For a cumulative event word $w_t = s_{\alpha_1} s_{\alpha_2} \cdots s_{\alpha_t}$, the corresponding composite surface homeomorphism is given by $\tilde{\Phi}(w_t) = \phi_{\alpha_1} \circ \phi_{\alpha_2} \circ \cdots \circ \phi_{\alpha_t}$, where $\phi_{\alpha_i} = \tilde{\Phi}(s_{\alpha_i})$. For an initial point $x_0 \in \Sigma$, the microscopic point-trajectory $(x_t)_{t \geq 0}$ under the event sequence $\mathbf{s}$ evolves by sequentially applying homeomorphisms in chronological order:

$$x_t = (\phi_{\alpha_t} \circ \cdots \circ \phi_{\alpha_2} \circ \phi_{\alpha_1})(x_0) = \tilde{\Phi}(s_{\alpha_t} \cdots s_{\alpha_2} s_{\alpha_1})(x_0).$$

In this formulation, the reversal of word order relative to function composition reflects the standard evaluation convention for maps acting on points from the left, ensuring complete algebraic and dynamic consistency without introducing contradictory right-action conventions.

Definition 3.3 (Sequence of Emergent Manifolds). Given a discrete micro-event dynamical system $(\Sigma, G, \rho, \tilde{\Phi}, \mathbf{s})$, the temporal trajectory of the system induces a dynamic sequence of 3-dimensional mapping tori $\{M_t\}_{t \geq 1}$, defined by

$$M_t := M_{\phi_{g_t}} = \frac{\Sigma \times [0, 1]}{(x, 0) \sim (\phi_{g_t}(x), 1)},$$

where $[\phi_{g_t}] = \rho(g_t) = \Phi(w_t) \in \text{MCG}(\Sigma)$.

Definition 3.4 (Path Dependence and Topological Non-Equivalence). Let $\mathbf{s}^A, \mathbf{s}^B$ be two micro-event streams generating accumulated words $g_t^A, g_t^B \in G$ at step t . The distinction between paths is classified along the following hierarchical levels of path dependence:

- (1) **Algebraic Path Dependence:** The paths are algebraically distinct if $g_t^A \neq g_t^B$ in G .
- (2) **Representation Path Dependence:** The paths are representationally distinct under $\rho : G \rightarrow \text{MCG}(\Sigma)$ if $\rho(g_t^A) \neq \rho(g_t^B)$ in $\text{MCG}(\Sigma)$, which requires $g_t^A (g_t^B)^{-1} \notin \ker \rho$.
- (3) **Topological Path Dependence (Non-isomorphic Mapping Tori):** The paths generate topologically non-equivalent 3-manifold invariants if the associated mapping tori are not homeomorphic:

$$M_{\rho(g_t^A)} \not\cong M_{\rho(g_t^B)}.$$

Note that if $\rho(g_t^A)$ and $\rho(g_t^B)$ are conjugate in $\text{MCG}(\Sigma)$ (up to orientation-reversing equivalence), then their mapping tori are homeomorphic ($M_{\rho(g_t^A)} \cong M_{\rho(g_t^B)}$).

Two trajectories g_t^A, g_t^B are said to exhibit **Strict Topological Path Dependence** if $M_{\rho(g_t^A)} \not\cong M_{\rho(g_t^B)}$.

Proposition 3.5 (Fundamental Group Presentation of Mapping Torus). *Let $w \in F(S)$ and $\phi_w = \Phi(w) \in \text{Homeo}^+(\Sigma)$. The fundamental group of the mapping torus M_{ϕ_w} is isomorphic to the semidirect product:*

$$\pi_1(M_{\phi_w}) \cong \pi_1(\Sigma) \rtimes_{(\phi_w)_*} \mathbb{Z}.$$

Explicitly, if $t \in \pi_1(M_{\phi_w})$ represents the stable generator of the S^1 -fiber loop, $\pi_1(M_{\phi_w})$ admits the finite presentation:

$$\pi_1(M_{\phi_w}) = \left\langle a_1, b_1, \dots, a_g, b_g, t \left| \prod_{i=1}^g [a_i, b_i] = 1, \quad txt^{-1} = (\phi_w)_*(x) \quad \forall x \in \{a_i, b_i\}_{i=1}^g \right. \right\rangle.$$

4. STOCHASTIC MICRO-DYNAMICS

To model realistic complex systems where micro-events occur non-deterministically, we extend our deterministic framework to a probabilistic setting. We formalize micro-event sequences as random walks on the finitely presented group $G = \langle S \mid R \rangle$ (and its covering free group $F(S)$), which generate a dynamic sequence of random mapping classes and random 3-dimensional mapping tori.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $S^{\pm 1} = S \cup S^{-1}$ denote the symmetric generating set of G .

Definition 4.1 (Stochastic Event Stream). A **stochastic event stream** is a sequence of $S^{\pm 1}$ -valued independent and identically distributed (i.i.d.) random variables

$$(X_n)_{n \geq 1}, \quad X_n : \Omega \longrightarrow S^{\pm 1},$$

governed by a probability measure μ on $S^{\pm 1}$. We assume that μ is non-degenerate (i.e., $\text{supp}(\mu) = S^{\pm 1}$) and symmetric ($\mu(s^{-1}) = \mu(s)$ for all $s \in S^{\pm 1}$).

Definition 4.2 (Random Word Trajectory and Group State). For each $n \in \mathbb{N}_0$, the n -th **cumulative random word** $W_n \in F(S)$ is defined inductively by

$$W_0 = e, \quad W_n = W_{n-1}X_n = X_1X_2 \cdots X_n.$$

The corresponding **random group state** G_n is given by the canonical projection $G_n = [W_n] \in G$, defining a symmetric random walk on G with initial state $G_0 = e$.

Definition 4.3 (Sequence of Random Mapping Tori). Let $\rho : G \rightarrow \text{MCG}(\Sigma)$ be a group representation. The sequence of random group states $(G_n)_{n \geq 0}$ induces a sequence of random mapping classes $[\phi_n] \in \text{MCG}(\Sigma)$ and a sequence of random 3-dimensional mapping tori $\{M_n\}_{n \geq 1}$ given by

$$[\phi_n] = \rho(G_n), \quad M_n = M_{\phi_n} = \frac{\Sigma \times [0, 1]}{(x, 0) \sim (\phi_n(x), 1)}.$$

The sequence $\{M_n\}_{n \geq 1}$ defines a discrete-time stochastic process taking values in the space $\mathcal{T}_{\text{MT}}(\Sigma)$ of homeomorphism classes of mapping tori.

Definition 4.4 (Filtration and Event History). Let $\mathcal{F}_n = \sigma(X_1, X_2, \dots, X_n)$ be the natural filtration generated by the micro-event stream up to time n , with $\mathcal{F}_0 = \{\emptyset, \Omega\}$. The sequence of cumulative random words $(W_n)_{n \geq 0}$, group states $(G_n)_{n \geq 0}$, and random mapping tori $\{M_n\}_{n \geq 0}$ are adapted stochastic processes with respect to $(\mathcal{F}_n)_{n \geq 0}$.

[Topological Action Order and Composition] Let $\text{MCG}(\Sigma)$ be the mapping class group. We define the representation $\rho : G \rightarrow \text{MCG}(\Sigma)$ as a group homomorphism satisfying:

$$\rho(gh) = \rho(g)\rho(h) \quad \forall g, h \in G.$$

For a cumulative random word trajectory $W_n = X_1 X_2 \cdots X_n$ with $X_i \in S^{\pm 1}$, the corresponding mapping class representation decomposes linearly with respect to the sequence of increments:

$$\rho(W_n) = \rho(X_1)\rho(X_2) \cdots \rho(X_n).$$

Under the shift transition operator $T_s(g) = gs$, this yields the exact time-ordered composition:

$$(T_{X_n} \circ \cdots \circ T_{X_1})(e) = X_1 X_2 \cdots X_n = W_n.$$

Proposition 4.5 (Markov Property of State Evolution). *The sequence of random group states $(G_n)_{n \geq 0}$ forms a homogeneous Markov chain on G with transition kernel $P(g, h) = \mu(g^{-1}h)$. Furthermore, while the sequence of random mapping tori $\{M_n\}_{n \geq 1}$ is not in general Markovian due to topological coarse-graining, the joint process $(G_n, M_n)_{n \geq 0}$ satisfies the Markov property relative to the filtration $(\mathcal{F}_n)_{n \geq 0}$.*

5. TOPOLOGICAL OBSERVABLES AND COARSE-GRAINING HIERARCHY

5.1. Topological Observables. Let $M_n = M_{\Phi(W_n)}$ be the random 3-manifold generated at step n by the mapping class $[\phi_n] = \rho(G_n) \in \text{MCG}(\Sigma)$.

Definition 5.1 (Topological Observable Vector). The macroscopic topological observable vector associated with the emergent mapping torus M_n is defined as the tuple:

$$\mathcal{O}(M_n) := (\mathbf{T}(\phi_n), h_{\text{top}}(\phi_n), H_1(M_n; \mathbb{Z}), \pi_1(M_n), \text{Vol}_{\text{hyp}}(M_n))$$

where each component is defined as follows:

- (1) $\mathbf{T}(\phi_n) \in \{\text{Periodic, Reducible, pseudo-Anosov}\}$ denotes the Nielsen–Thurston topological type of the mapping class $[\phi_n]$;
- (2) $h_{\text{top}}(\phi_n) \geq 0$ is the topological entropy of the induced mapping class $[\phi_n]$ (formalized in Section 8);
- (3) $H_1(M_n; \mathbb{Z})$ is the first integral homology group of the mapping torus M_n ;
- (4) $\pi_1(M_n) \cong \pi_1(\Sigma) \rtimes_{(\phi_n)_*} \mathbb{Z}$ is the fundamental group of M_n ;
- (5) $\text{Vol}_{\text{hyp}}(M_n)$ denotes the hyperbolic volume of M_n , defined piecewisely by:

$$\text{Vol}_{\text{hyp}}(M_n) = \begin{cases} \text{Vol}(M_n, g_{\text{hyp}}), & \text{if } [\phi_n] \text{ is pseudo-Anosov (so } M_n \text{ is hyperbolic),} \\ 0, & \text{otherwise.} \end{cases}$$

Remark 5.2 (Well-Definedness of Observables). By Thurston’s Hyperbolization Theorem, M_n admits a unique complete, finite-volume hyperbolic metric g_{hyp} if and only if $[\phi_n]$ is pseudo-Anosov. Assigning $\text{Vol}_{\text{hyp}}(M_n) = 0$ for non-hyperbolic mapping tori guarantees that $\mathcal{O}(M_n)$ is globally well-defined across all Nielsen–Thurston types without domain restrictions.

5.2. Coarse-Graining Hierarchy and Information Loss.

Definition 5.3 (Hierarchical Equivalence Relations). For words $w_1, w_2 \in F(S)$, we define:

- (1) **Micro-event identity** $w_1 = w_2$ in $F(S)$.
- (2) **Algebraic equivalence** $(w_1 \sim_G w_2)$: $[w_1] = [w_2] \in G$.
- (3) **Mapping class equivalence** $(w_1 \sim_{\text{MCG}} w_2)$: $\rho([w_1]) = \rho([w_2]) \in \text{MCG}(\Sigma)$.
- (4) **Topological equivalence** $(w_1 \sim_{\text{top}} w_2)$: $M_{\Phi(w_1)} \cong M_{\Phi(w_2)}$.

Theorem 5.4 (Inclusion Hierarchy). *The equivalence relations satisfy the nested inclusion chain:*

$$\boxed{\sim_G \subseteq \sim_{\text{MCG}} \subseteq \sim_{\text{top}}}$$

Proof. If $w_1 \sim_G w_2$, then $[w_1] = [w_2] \in G$. Applying the representation homomorphism ρ , we obtain $\rho([w_1]) = \rho([w_2])$, which implies $w_1 \sim_{\text{MCG}} w_2$. Next, if $w_1 \sim_{\text{MCG}} w_2$, then $\rho([w_1]) = \rho([w_2]) = [\phi] \in \text{MCG}(\Sigma)$. By Proposition ?? (Invariance Under Isotopy), isotopic representative homeomorphisms yield homeomorphic mapping tori $M_{\Phi(w_1)} \cong M_{\Phi(w_2)}$, establishing $w_1 \sim_{\text{top}} w_2$. $\square$

Remark 5.5 (Conditions for Strict Inclusions). The inclusion chain $\sim_G \subseteq \sim_{\text{MCG}} \subseteq \sim_{\text{top}}$ is weak in general.

- (1) If the representation $\rho : G \rightarrow \text{MCG}(\Sigma)$ is injective, then $\sim_G = \sim_{\text{MCG}}$, meaning no algebraic information is lost at the representation level. Conversely, if $\ker \rho \neq \{e\}$, the first inclusion becomes strict ($\sim_G \subsetneq \sim_{\text{MCG}}$).
- (2) Similarly, the second inclusion becomes strict ($\sim_{\text{MCG}} \subsetneq \sim_{\text{top}}$) whenever distinct (non-isotopic) mapping classes generate homeomorphic mapping tori (e.g., via conjugate mapping classes, as demonstrated in Section 11).

6. NON-COMMUTATIVE DYNAMICS ON MANIFOLDS

The preceding sections constructed the algebraic, stochastic, and topological components of the proposed framework. We now combine these components to define and analyze a dynamical system whose state is induced by an ordered sequence of non-commutative events and represented geometrically by a family of manifolds.

Let $W_n = X_1 X_2 \cdots X_n$ denote the ordered product of elementary events, and let $G_n = [W_n] \in G$ be the corresponding algebraic state. Through the representation

$$\rho : G \longrightarrow \text{MCG}(\Sigma),$$

the state G_n induces a mapping class $[\phi_n] = \rho(G_n)$, and hence a mapping torus $M_n = M_{\phi_n}$. The resulting sequence

$$M_0, M_1, M_2, \dots$$

defines the geometric state evolution generated by the microscopic event process.

The central object of this section is therefore not non-commutativity alone, but the dynamical behavior induced by non-commutative evolution on the associated manifolds. We investigate how the ordering of elementary events affects the evolution of the algebraic state, the mapping class, and the resulting manifold, and how these effects are reflected in topological and geometric observables. In particular, we study the dynamic chain

$$W_n \longrightarrow G_n \longrightarrow \rho(G_n) \longrightarrow M_n \longrightarrow \mathcal{O}(M_n),$$

and analyze the resulting behavior at each level.

6.1. State Space and Induced Manifold Dynamics. To formalize the dynamic evolution generated by micro-events, we clarify the nature of the underlying state space. Rather than a classical dynamical system defined by an iterated map $x_{n+1} = f(x_n)$ on a single, fixed background manifold M , our construction defines a **state-dependent dynamic manifold system**, where the underlying geometric space itself evolves as a function of cumulative non-commutative event history.

Definition 6.1 (Reachable State Space). Let Σ be a reference surface, and let $\psi : \text{MCG}(\Sigma) \rightarrow \mathcal{T}_{\text{MT}}(\Sigma)$ be the canonical topology assignment operator mapping each mapping class ϕ to the homeomorphism class of its mapping torus $[M_\phi]$.

Given a representation $\rho : G \rightarrow \text{MCG}(\Sigma)$, the reachable state space $\mathcal{X}_\rho$ of the induced dynamical system is defined as the graph subspace of the ambient product space $G \times \text{MCG}(\Sigma) \times \mathcal{T}_{\text{MT}}(\Sigma)$:

$$\mathcal{X}_\rho := \{(g, \rho(g), \psi(\rho(g))) \in G \times \text{MCG}(\Sigma) \times \mathcal{T}_{\text{MT}}(\Sigma) \mid g \in G\}.$$

Definition 6.2 (Manifold Transition Operator). Let $S = \{s_1, \dots, s_k\}$ be the generating set of G . The micro-event dynamics on $\mathcal{X}_\rho$ is driven by the state-transition operator $T_s : \mathcal{X}_\rho \rightarrow \mathcal{X}_\rho$ parameterized by $s \in S^{\pm 1}$, defined via right-multiplication:

$$T_s(g, \rho(g), \psi(\rho(g))) = (gs, \rho(gs), \psi(\rho(gs))).$$

For an ordered sequence of event generators $W_n = X_1 X_2 \cdots X_n$ (where $X_i \in S^{\pm 1}$), the discrete trajectory starting from the identity state $x_0 = (e, \text{id}, [M_{\text{id}}])$ satisfies:

$$(T_{X_n} \circ \cdots \circ T_{X_1})(x_0) = (W_n, \rho(W_n), [M_n]),$$

where the sequence of induced manifolds $\{M_n\}_{n \geq 0}$ satisfies $[M_n] = \psi(\rho(W_n)) = \psi(\rho(X_1) \circ \cdots \circ \rho(X_n))$.

Remark 6.3 (Microscopic vs. Macroscopic Dynamics). This formulation explicitly separates two complementary dynamical levels:

- (1) **Microscopic Trajectory:** Point dynamics $x_n = \phi_{X_n}(x_{n-1})$ acting locally on the reference surface Σ ;
- (2) **Macroscopic Trajectory:** Structural manifold evolution $M_{n-1} \mapsto M_n$ defined by the changing global topology of the mapping torus.

6.2. Criteria for Detecting Non-Commutative Path Dependence. Let $g \in G$ be an arbitrary base state, and let $a, b \in S$ be non-commuting generators ($ab \neq ba$ in G). Consider two trajectory branches starting from g after two time steps:

$$g_A = g \cdot (ab) \quad \text{and} \quad g_B = g \cdot (ba).$$

Proposition 6.4 (Criteria for Detecting Non-Commutative Path Dependence). *Let $\rho : G \rightarrow \text{MCG}(\Sigma)$ be a group representation, and let $M = \sigma \circ \rho : G \rightarrow \text{Sp}(2g, \mathbb{Z})$ denote the symplectic homology representation.*

- (1) **MCG-Level Divergence:** *If the commutator $[\rho(a), \rho(b)] \neq \text{id}$ in $\text{MCG}(\Sigma)$, then*

$$\rho(g_A) = \rho(g)\rho(a)\rho(b) \neq \rho(g)\rho(b)\rho(a) = \rho(g_B).$$

- (2) **Homological Matrix Divergence:** *If $[M(a), M(b)] \neq I$ in $\text{Sp}(2g, \mathbb{Z})$, then*

$$M(g_A) = M(g)M(a)M(b) \neq M(g)M(b)M(a) = M(g_B).$$

- (3) **H_1 Homology Distinction:** *If the Smith Normal Forms satisfy*

$$\text{SNF}(M(g_A) - I) \neq \text{SNF}(M(g_B) - I),$$

then the first homology groups of the mapping tori are non-isomorphic:

$$H_1(M_{g_A}; \mathbb{Z}) \not\cong H_1(M_{g_B}; \mathbb{Z}).$$

Proof. For part (1), since $[\rho(a), \rho(b)] \neq \text{id}$, we have $\rho(a)\rho(b) \neq \rho(b)\rho(a)$. Multiplying by $\rho(g)$ on the left yields $\rho(g_A) \neq \rho(g_B)$. No global faithfulness assumption on ρ is required beyond non-triviality of the commutator $[\rho(a), \rho(b)]$.

Part (2) follows from the representation property of $M = \sigma \circ \rho$. Given $M(a)M(b) \neq M(b)M(a)$, multiplying by $M(g)$ directly yields $M(g_A) = M(g)M(a)M(b) \neq M(g)M(b)M(a) = M(g_B)$.

For part (3), recall that $H_1(M_\phi; \mathbb{Z}) \cong \mathbb{Z} \oplus \text{cok}(M(\phi) - I)$. The abelian group $\text{cok}(M(\phi) - I)$ is uniquely determined up to isomorphism by the diagonal entries of the Smith Normal Form of $M(\phi) - I$. Thus, $\text{SNF}(M(g_A) - I) \neq \text{SNF}(M(g_B) - I)$ implies $\text{cok}(M(g_A) - I) \not\cong \text{cok}(M(g_B) - I)$, which completes the proof. $\square$

6.3. Dynamical Observables and Growth Equations. To track the macroscopic state of the manifold sequence $\{M_n\}_{n \geq 0}$, we define temporal series of topological and geometric observables.

Definition 6.5 (Observable Trajectory Vectors). Along a dynamic manifold sequence $\{M_n\}_{n \geq 0}$, the sequence of observable vectors $\mathcal{O}(M_n) = (h_n, \Lambda_n, V_n)$ is defined by:

$$h_n := h_{\text{top}}(\rho(G_n)), \quad \Lambda_n := e^{h_n}, \quad V_n := \text{Vol}_{\text{hyp}}(M_n),$$

where Λ_n represents the exponential expansion factor of the induced mapping class ($\Lambda_n = \lambda(\rho(G_n))$) if $\rho(G_n)$ is pseudo-Anosov with dilatation λ , and $\Lambda_n = 1$ if $h_n = 0$).

Proposition 6.6 (Entropy-Translation Relation and Expected Entropy Rate). *Let $\tau_{\mathcal{T}}(\phi) := \inf_{x \in \mathcal{T}(\Sigma)} d_{\mathcal{T}}(x, \phi x)$ denote the Teichmüller translation length of a mapping class $\phi \in \text{MCG}(\Sigma)$.*

- (1) **Pseudo-Anosov Metric Identification:** *For any pseudo-Anosov mapping class $\phi \in \text{MCG}(\Sigma)$ with dilatation $\lambda(\phi) > 1$, the Teichmüller translation length coincides exactly with the topological entropy:*

$$\tau_{\mathcal{T}}(\phi) = \log \lambda(\phi) = h_{\text{top}}(\phi).$$

- (2) **Mean Asymptotic Entropy Rate:** *For a random walk trajectory $W_n = X_1 X_2 \cdots X_n$ on G governed by a step distribution μ whose image subgroup $G_\mu = \langle \rho(\text{supp } \mu) \rangle \subseteq \text{MCG}(\Sigma)$ is non-elementary, the expected topological entropy growth rate converges to a well-defined positive limit:*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}[h_{\text{top}}(\rho(W_n))] = L_\mu(\rho) > 0.$$

Proof. The identity $h_{\text{top}}(\phi) = \tau_{\mathcal{T}}(\phi) = \log \lambda(\phi)$ for pseudo-Anosov elements is a classical result of Fathi and Shub.

To establish the convergence of the asymptotic entropy growth rate, fix a basepoint $x_0 \in \mathcal{T}(\Sigma)$. The metric displacement process $D_n := d_{\mathcal{T}}(x_0, \rho(W_n)x_0)$ forms a subadditive stochastic process under functional composition on $(\mathcal{T}(\Sigma), d_{\mathcal{T}})$. By Kingman's Subadditive Ergodic Theorem, the limit

$$L_\mu(\rho) := \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}[d_{\mathcal{T}}(x_0, \rho(W_n)x_0)]$$

exists and is finite.

Furthermore, since G_μ is a non-elementary subgroup of $\text{MCG}(\Sigma)$, generic random walk paths hit pseudo-Anosov elements with exponentially high probability (Maher), and the walk escapes at a positive linear drift rate along Teichmüller geodesics (Gouëzel–Mathieu–Karlsson / Sisto). On the generic pseudo-Anosov sequence, the relation $h_{\text{top}}(\rho(W_n)) = \tau_{\mathcal{T}}(\rho(W_n)) = d_{\mathcal{T}}(x_0, \rho(W_n)x_0) + O(1)$ holds almost surely, proving that $\lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}[h_{\text{top}}(\rho(W_n))] = L_\mu(\rho) > 0$. $\square$

6.4. Algebraic, Topological, and Entropic Complexity Growth. We quantify the relationship between algebraic word expansion on the Cayley graph $\Gamma(G, S)$ and the topological entropy growth of the manifold family.

Theorem 6.7 (Entropic Growth Transfer Formula). *Let $(G_n)_{n \geq 0}$ be a non-degenerate random walk on $G = \langle S \mid R \rangle$ with drift velocity $v = \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}[d_S(e, G_n)] > 0$. Let $\rho : G \rightarrow \text{MCG}(\Sigma)$ be a non-elementary representation with asymptotic Lyapunov exponent $L_\mu(\rho) > 0$.*

Then the entropic efficiency ratio $\eta_{\text{top}} := \frac{L_\mu(\rho)}{v}$ measures the asymptotic topological entropy generated per unit word distance in the Cayley graph:

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}[h_{\text{top}}(\rho(G_n))]}{\mathbb{E}[d_S(e, G_n)]} = \eta_{\text{top}} = \frac{L_\mu(\rho)}{v}.$$

Proof. By the random walk drift property, $\mathbb{E}[d_S(e, G_n)] = vn + o(n)$ with $v > 0$. By subadditive ergodic theory applied to mapping class representations, $\mathbb{E}[h_{\text{top}}(\rho(G_n))] = L_\mu(\rho)n + o(n)$ with $L_\mu(\rho) > 0$. Taking the ratio of both asymptotic expressions as $n \rightarrow \infty$ yields:

$$\lim_{n \rightarrow \infty} \frac{L_\mu(\rho)n + o(n)}{vn + o(n)} = \frac{L_\mu(\rho)}{v} = \eta_{\text{top}},$$

completing the proof. $\square$

6.5. Long-Term Dynamical Behavior and Regime Structure. Finally, we classify the long-term asymptotic behavior of the dynamic manifold system $\{M_n\}_{n \geq 0}$ into distinct dynamical regimes governed by the micro-event step measure μ_p .

Theorem 6.8 (Regime Classification of Manifold Trajectories). *Let $p \in [0, 1]$ parameterize a family of step distributions μ_p . The long-term dynamical behavior of $\{M_n\}_{n \geq 1}$ partitions into three distinct macroscopic regimes:*

- (1) **Integrable / Periodic Regime** ($p = 0$): $\rho(G_0)$ contains no pseudo-Anosov elements (e.g., $\rho(G_0)$ is finite or periodic). $h_n = 0$ for all n , $\text{Vol}_{\text{hyp}}(M_n) = 0$ globally, and M_n remains Seifert-fibered or Euclidean torus bundles for all n .
- (2) **Sub-Hyperbolic / Reducible Regime** ($0 < p < p_c$ **with structural constraints**): $\rho(G)$ preserves a finite system of essential curves on Σ . Entropy growth is confined to lower-genus sub-surfaces, $\text{Vol}_{\text{hyp}}(M_n) = 0$ globally, and M_n exhibits graph-manifold decompositions.
- (3) **Generic Hyperbolic / Pseudo-Anosov Regime** ($p > 0$ **under non-elementary support**): $\rho(G_p)$ is non-elementary. As $n \rightarrow \infty$, M_n is pseudo-Anosov with probability 1, exhibiting mean asymptotic entropy growth $\frac{1}{n}\mathbb{E}[h_n] \rightarrow L_\mu(\rho) > 0$ and asymptotically positive hyperbolic volume growth.

6.6. Illustrative Example: Micro-Path Divergence vs. Topological Coarse-Graining.

To illustrate the relationship between non-commutative event orderings and manifold topology, we examine an explicit example demonstrating how micro-level algebraic divergence interacts with macro-level topological equivalence.

Let $a, b \in S$ be two non-commuting generators whose images $\rho(a), \rho(b) \in \text{MCG}(\Sigma)$ do not commute ($[\rho(a), \rho(b)] \neq \text{id}$). Consider two word sequences generated by swapping the order of these two steps:

$$w_A = ab \quad \text{and} \quad w_B = ba.$$

At the algebraic and mapping class level, non-commutativity strictly separates the micro-states:

$$w_A \neq w_B \in G \implies \rho(w_A) = \rho(b) \circ \rho(a) \neq \rho(a) \circ \rho(b) = \rho(w_B).$$

However, observe the algebraic identity in G :

$$w_B = ba = a^{-1}(ab)a = a^{-1}w_Aa.$$

Applying the representation ρ , we obtain:

$$\rho(w_B) = \rho(a)^{-1} \circ \rho(w_A) \circ \rho(a).$$

Thus, $\rho(w_A)$ and $\rho(w_B)$ are ****conjugate elements**** in $\text{MCG}(\Sigma)$. By standard 3-manifold topology, conjugate mapping classes induce homeomorphic mapping tori:

$$M_{\rho(w_A)} \cong M_{\rho(w_B)}.$$

This example provides crucial physical and mathematical insight: non-commutative path divergence at the algebraic micro-level ($w_A \neq w_B$, $\rho(w_A) \neq \rho(w_B)$) does not necessarily imply non-isomorphic global geometry ($M_{\rho(w_A)} \cong M_{\rho(w_B)}$). The transition $M_{\rho(w_n)} \rightarrow \mathcal{O}(M_n)$ represents a

****topological coarse-graining****, where micro-level path dependence can be quotiented out by the conjugacy classes of the mapping class group.

7. MICROSCOPIC, ALGEBRAIC, AND TOPOLOGICAL COMPLEXITY GROWTH

Having defined the observables and coarse-graining hierarchy, we quantify how microscopic non-commutativity scales across algebraic and topological state spaces. We formalize three distinct notions of complexity for a cumulative random word trajectory $W_n = X_1 X_2 \cdots X_n$.

Definition 7.1 (Three Complexity Invariants). For a cumulative random word $W_n \in F(S)$ at step n , we define:

- (1) **Microscopic Word Complexity:** $C_{\text{micro}}(W_n) := |W_n| = n$, representing the raw physical sequence length.
- (2) **Algebraic Complexity:** $C_{\text{alg}}(W_n) := d_S(e, [W_n])$, the word metric distance from the identity on the Cayley graph $\Gamma(G, S)$.
- (3) **Topological Complexity:** $C_{\text{top}}(W_n) := h_{\text{top}}(\rho([W_n]))$, the topological entropy of the induced mapping class in $\text{MCG}(\Sigma)$.

Theorem 7.2 (Independent Asymptotic Growth Rates). *Let Σ be a closed, connected, orientable surface of genus $g \geq 2$, and let $G = \langle S \mid R \rangle$ be a group. Let $\rho : G \rightarrow \text{MCG}(\Sigma)$ be a representation such that the image subgroup $\rho(G) \subseteq \text{MCG}(\Sigma)$ is non-elementary. Then:*

- (1) **Algebraic Complexity Rate:** *The expected algebraic complexity satisfies*

$$\mathbb{E}[C_{\text{alg}}(W_n)] = v \cdot n + o(n) \quad \text{as } n \rightarrow \infty,$$

where $v = \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}[d_S(e, G_n)] \geq 0$ is the asymptotic escape rate (drift) on G .

- (2) **Topological Complexity Rate:** *The expected topological complexity satisfies*

$$\mathbb{E}[C_{\text{top}}(W_n)] = L_\mu(\rho) \cdot n + o(n) \quad \text{as } n \rightarrow \infty,$$

where $L_\mu(\rho) > 0$ is the asymptotic Lyapunov exponent in $\text{MCG}(\Sigma)$.

Corollary 7.3 (Asymptotic Complexity Growth Ratio). *By Theorem 7.2, if the random walk on G exhibits positive drift ($v > 0$), then the ratio of expected topological complexity to expected algebraic complexity satisfies:*

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}[C_{\text{top}}(W_n)]}{\mathbb{E}[C_{\text{alg}}(W_n)]} = \frac{L_\mu(\rho)}{v}.$$

Proof. Assume $v > 0$. Dividing the asymptotic expansions from Theorem 7.1 by n , we obtain:

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n} \mathbb{E}[C_{\text{top}}(W_n)]}{\frac{1}{n} \mathbb{E}[C_{\text{alg}}(W_n)]} = \frac{L_\mu(\rho) + o(1)}{v + o(1)} = \frac{L_\mu(\rho)}{v}.$$

This completes the proof. □

8. PUSH-FORWARD STOCHASTIC DYNAMICS

Having defined the pipeline, we formalize the stochastic evolution of probability measures driven by micro-event random walks through the state spaces.

Definition 8.1 (Group Random Walk and Transition Kernel). Let μ be a step probability distribution on the symmetric generating set $S^{\pm 1}$ of G . The sequence of random group states $(G_n)_{n \geq 0}$ forms a Markov chain on G with initial state $G_0 = e$ and transition probability:

$$P(G_{n+1} = gs \mid G_n = g) = \mu(s), \quad \forall g \in G, s \in S^{\pm 1}.$$

The n -step distribution μ^{*n} on G is given by n -fold convolution: $\mu^{*n}(g) = \mathbb{P}(G_n = g)$.

Definition 8.2 (Push-Forward Measure Pipeline). The probability measure μ on micro-events induces a cascade of push-forward measures across the algebraic, mapping class, topological, and observable spaces:

$$\mu \longrightarrow \mu_{\text{alg}}^{(n)} := \pi_*(\mu^{*n}) \longrightarrow \mu_{\text{MCG}}^{(n)} := \rho_*(\mu_{\text{alg}}^{(n)}) \longrightarrow \mu_{\text{top}}^{(n)} := \text{MT}_*(\mu_{\text{MCG}}^{(n)}) \longrightarrow \mu_{\text{obs}}^{(n)} := \mathcal{O}_*(\mu_{\text{top}}^{(n)})$$

where:

- (1) $\mu_{\text{alg}}^{(n)}$ is the probability distribution on G at step n ;
- (2) $\mu_{\text{MCG}}^{(n)}$ is the pushed-forward measure on $\text{MCG}(\Sigma)$;
- (3) $\mu_{\text{top}}^{(n)}$ is the measure on the space of mapping tori $\mathcal{T}_{\text{MT}}(\Sigma)$;
- (4) $\mu_{\text{obs}}^{(n)}$ is the push-forward measure on the observable vector space $\mathbb{R}^k \times \text{Groups}$.

9. TOPOLOGICAL COMPLEXITY AND ENTROPY GROWTH

Definition 9.1 (Topological Entropy of Mapping Classes). Let $[\phi] \in \text{MCG}(\Sigma)$ ($g \geq 2$) reduce along canonical curves into irreducible components $\{\Sigma_i\}_{i=1}^m$. The topological entropy $h_{\text{top}}([\phi])$ is defined as:

$$h_{\text{top}}([\phi]) = \max_{1 \leq i \leq m} h_{\text{top}}(\phi|_{\Sigma_i}) = \max_{i \in \text{pA}} \log \lambda_i,$$

where $\lambda_i > 1$ denotes the dilatation of the pseudo-Anosov piece $\phi|_{\Sigma_i}$. If $[\phi]$ contains no pseudo-Anosov components, $h_{\text{top}}([\phi]) = 0$.

Definition 9.2 (Asymptotic Entropy Growth Rate). The asymptotic entropy growth rate $L_\mu(\rho)$ is defined by:

$$L_\mu(\rho) := \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}_{\mathbb{P}} [h_{\text{top}}(\rho(W_n))].$$

9.1. Non-Vanishing Asymptotic Entropy Growth.

Definition 9.3 (Non-Elementary Subgroup of $\text{MCG}(\Sigma)$). A subgroup $H \subseteq \text{MCG}(\Sigma)$ is called **non-elementary** if H does not fix any finite set of essential simple closed curves on Σ (up to isotopy) and does not fix a finite set of points on the Thurston boundary $\mathcal{PMF}(\Sigma)$. Equivalently, H contains at least two pseudo-Anosov mapping classes with disjoint fixed point pairs on $\mathcal{PMF}(\Sigma)$.

Theorem 9.4 (Positivity of Asymptotic Entropy Growth Rate). *Let Σ be a closed, connected, orientable surface of genus $g \geq 2$, and let $G = \langle S \mid R \rangle$ be a finitely presented group with representation $\rho : G \rightarrow \text{MCG}(\Sigma)$. Let μ be a non-degenerate probability measure on $S^{\pm 1}$ with finite support, and let $W_n = X_1 X_2 \cdots X_n$ denote the random walk on G generated by μ .*

If the induced image subgroup $G_\mu := \langle \rho(\text{supp } \mu) \rangle \subseteq \text{MCG}(\Sigma)$ is a non-elementary subgroup containing independent pseudo-Anosov elements, then by subadditive ergodic theory and Teichmüller geodesic drift, the expected topological entropy growth rate exists and is strictly positive:

$$L_\mu(\rho) := \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}_{\mathbb{P}} [h_{\text{top}}(\rho(W_n))] = \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}_{\mathbb{P}} [\tau_{\mathcal{T}}(\rho(W_n))] > 0.$$

Proof. Let $\mathcal{T}(\Sigma)$ denote the Teichmüller space of Σ equipped with the Teichmüller metric $d_{\mathcal{T}}$. Since G_μ is non-elementary, Maher's Genericity Theorem for random walks on mapping class groups applies: the probability that $\rho(W_n)$ is a pseudo-Anosov mapping class approaches 1 with exponential speed. Specifically, there exist constants $C, c > 0$ such that

$$\mathbb{P}(\mathcal{A}_n) \geq 1 - Ce^{-cn}, \quad \text{where } \mathcal{A}_n = \{\omega \in \Omega \mid \rho(W_n(\omega)) \text{ is pseudo-Anosov}\}.$$

By Proposition 6.6, for any sample path where $\rho(W_n)$ is pseudo-Anosov, its topological entropy coincides exactly with its Teichmüller translation length:

$$h_{\text{top}}(\rho(W_n)) = \log \lambda(\rho(W_n)) = \tau_{\mathcal{T}}(\rho(W_n)) \quad \text{on } \mathcal{A}_n.$$

Furthermore, by the drift rate theorem for isometric random walks on weakly hyperbolic metric spaces (Gouëzel–Mathieu–Karlsson and Sisto), the non-elementarity of G_μ guarantees a strictly positive asymptotic drift rate (escape rate) $l_\mu(\rho) > 0$ on $\mathcal{T}(\Sigma)$:

$$l_\mu(\rho) := \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}_{\mathbb{P}} [\tau_{\mathcal{T}}(\rho(W_n))] > 0.$$

We decompose the expectation of topological entropy over the event $\mathcal{A}_n$ and its complement $\mathcal{A}_n^c$:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}} [h_{\text{top}}(\rho(W_n))] &\geq \mathbb{E}_{\mathbb{P}} [h_{\text{top}}(\rho(W_n)) \mathbf{1}_{\mathcal{A}_n}] \\ &= \mathbb{E}_{\mathbb{P}} [\tau_{\mathcal{T}}(\rho(W_n)) \mathbf{1}_{\mathcal{A}_n}] \\ &= \mathbb{E}_{\mathbb{P}} [\tau_{\mathcal{T}}(\rho(W_n))] - \mathbb{E}_{\mathbb{P}} [\tau_{\mathcal{T}}(\rho(W_n)) \mathbf{1}_{\mathcal{A}_n^c}]. \end{aligned}$$

Let $K = \max_{s \in \text{supp } \mu} \tau_{\mathcal{T}}(\rho(s)) < \infty$. By the triangle inequality, $\tau_{\mathcal{T}}(\rho(W_n)) \leq Kn$. Thus, the error term bounded over non-pseudo-Anosov trajectories satisfies:

$$\mathbb{E}_{\mathbb{P}} [\tau_{\mathcal{T}}(\rho(W_n)) \mathbf{1}_{\mathcal{A}_n^c}] \leq Kn \cdot \mathbb{P}(\mathcal{A}_n^c) \leq CKne^{-cn}.$$

Dividing by n and taking the limit $n \rightarrow \infty$, the error term vanishes:

$$\lim_{n \rightarrow \infty} \frac{CKne^{-cn}}{n} = 0.$$

Therefore, we obtain the exact equality of the limits:

$$L_\mu(\rho) = \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}_{\mathbb{P}} [h_{\text{top}}(\rho(W_n))] = \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}_{\mathbb{P}} [\tau_{\mathcal{T}}(\rho(W_n))] = l_\mu(\rho) > 0,$$

which establishes that the asymptotic entropy growth rate is well-defined and strictly positive. $\square$

10. THURSTON CLASSIFICATION AND TOPOLOGICAL PHASE TRANSITIONS

Theorem 10.1 (Nielsen–Thurston Classification). *Let Σ be a closed, connected, orientable surface of genus $g \geq 2$. Every mapping class $[\phi] \in \text{MCG}(\Sigma)$ belongs to exactly one of three canonical types:*

- (1) **Periodic:** $[\phi]^k = \text{id}$ for some $k \in \mathbb{N}$;
- (2) **Reducible:** $[\phi]$ preserves a non-empty system of essential disjoint simple closed curves on Σ ;
- (3) **pseudo-Anosov:** $[\phi]$ preserves a pair of transverse singular measured foliations $(\mathcal{F}^s, \mu_s)$ and $(\mathcal{F}^u, \mu_u)$ with dilatation $\lambda > 1$.

Theorem 10.2 (Thurston’s Hyperbolization Theorem). *Let $[\phi] \in \text{MCG}(\Sigma)$ ($g \geq 2$). The mapping torus M_ϕ admits a complete, finite-volume hyperbolic metric if and only if $[\phi]$ is pseudo-Anosov.*

Definition 10.3 (Parametric Step Measure and Support Hypotheses). Let $p \in [0, 1]$ parameterize a family of symmetric step distributions μ_p on $S^{\pm 1}$. We assume the following explicit support conditions on the representation image subgroup $G_p := \langle \rho(\text{supp}(\mu_p)) \rangle \subseteq \text{MCG}(\Sigma)$:

- (1) **Commutative/Non-Pseudo-Anosov Regime** ($p = 0$): The support $\text{supp}(\mu_0)$ generates a subgroup $G_0 \subset \text{MCG}(\Sigma)$ consisting entirely of non-pseudo-Anosov elements (i.e., every element in G_0 is either periodic or reducible, containing no pseudo-Anosov mapping classes).
- (2) **Non-Elementary Regime** ($p > 0$): For any $p > 0$, the support $\text{supp}(\mu_p)$ contains sufficient non-commutative steps such that $G_p \subseteq \text{MCG}(\Sigma)$ is a non-elementary subgroup (i.e., G_p is not virtually cyclic and does not fix any finite set of essential curves or measured foliations on Σ).

Definition 10.4 (Asymptotic Hyperbolic Order Parameter). For $p \in [0, 1]$, we define the **asymptotic hyperbolic order parameter** $\Theta(p)$ as the limit probability that a length- n random word trajectory generates a pseudo-Anosov mapping class:

$$\Theta(p) := \lim_{n \rightarrow \infty} \mathbb{P}_{\mu_p}(\{\omega \in \Omega \mid \rho(W_n(\omega)) \text{ is pseudo-Anosov}\}).$$

Theorem 10.5 (Asymptotic Topological Phase Transition). *Let Σ be a surface of genus $g \geq 2$, and let μ_p ($p \in [0, 1]$) be the parameterized step distribution satisfying the Support Hypotheses. The order parameter $\Theta(p)$ exhibits a sharp topological phase transition at $p_c = 0$:*

$$\Theta(p) = \begin{cases} 0, & \text{if } p = 0 \quad (\text{Commutative / Non-Hyperbolic Phase}), \\ 1, & \text{if } p > 0 \quad (\text{Hyperbolic Phase}). \end{cases}$$

Consequently, for any $p > 0$, the random mapping torus M_n becomes asymptotically hyperbolic with probability 1 as $n \rightarrow \infty$, whereas for $p = 0$, $\text{Vol}_{\text{hyp}}(M_n) = 0$ almost surely for all n .

Proof. For $p = 0$: By definition, the subgroup $G_0 = \langle \text{supp}(\mu_0) \rangle$ consists entirely of non-pseudo-Anosov mapping classes (periodic or reducible elements). Since any finite product of elements in G_0 remains within G_0 , every realization of the random word W_n under μ_0 maps to a non-pseudo-Anosov class. Thus $\mathbb{P}_{\mu_0}(\rho(W_n) \text{ is pseudo-Anosov}) = 0$ for all n , yielding $\Theta(0) = 0$.

For $p > 0$: Since G_p is non-elementary, Maher's Genericity Theorem for random walks on mapping class groups applies. The probability that a random walk of length n yields a pseudo-Anosov element satisfies:

$$\mathbb{P}_{\mu_p}(\rho(W_n) \text{ is pseudo-Anosov}) \geq 1 - Ce^{-cn}$$

for positive constants $C, c > 0$ depending on p . Taking the limit $n \rightarrow \infty$ gives $\Theta(p) = 1$. By Thurston's Hyperbolization Theorem, M_n is hyperbolic if and only if $\rho(W_n)$ is pseudo-Anosov, establishing the asymptotic phase transition. $\square$

Remark 10.6 (Thermodynamic Interpretation of $\Theta(p)$). The step function $\Theta(p)$ serves as a rigorous geometric order parameter. The jump from $\Theta(0) = 0$ to $\Theta(p) = 1$ for any $p > 0$ proves that introducing even an infinitesimally small probability $p > 0$ of non-elementary, non-commutative micro-events radically changes the macroscopic asymptotic geometry of the state space manifold from Seifert-fibered/flat structures to hyperbolic geometry.

11. CONCRETE EXAMPLE: TOPOLOGICAL COARSE-GRAINING AND CONJUGACY ON Σ_2

To illustrate the operational mechanism of topological coarse-graining and micro-path dependence, we construct an explicit structural example on a reference surface of genus 2, Σ_2 .

11.1. Generator Construction and Event Orderings. Let $S = \{s_1, s_2\}$ be the generating set for micro-events, and let $\rho : F_2 \rightarrow \text{MCG}(\Sigma_2)$ be a group representation mapping generators s_1, s_2 to mapping classes $[\phi_1], [\phi_2] \in \text{MCG}(\Sigma_2)$. Consider two micro-event word sequences generated by swapping the temporal ordering of two elementary steps:

$$w_A = s_1 s_2 \quad \text{and} \quad w_B = s_2 s_1.$$

Proposition 11.1 (Conjugacy Invariance and Topological Coarse-Graining). *Let $S = \{s_1, s_2\}$ be the generating set for micro-events, and let $\rho : F_2 \rightarrow \text{MCG}(\Sigma_2)$ be a group representation mapping generators s_1, s_2 to mapping classes $[\phi_1], [\phi_2] \in \text{MCG}(\Sigma_2)$ satisfying $[\rho(s_1), \rho(s_2)] \neq \text{id}$ in $\text{MCG}(\Sigma_2)$. Under this representation:*

- (1) **Microscopic Path Divergence:** *Because $[\rho(s_1), \rho(s_2)] \neq \text{id}$, the induced mapping classes satisfy $\rho(w_A) = \rho(s_1)\rho(s_2) \neq \rho(s_2)\rho(s_1) = \rho(w_B)$.*

- (2) **Conjugacy Relation:** The word sequences satisfy the algebraic conjugacy relation $w_B = s_1^{-1}w_A s_1$ in F_2 . Consequently, their image mapping classes are conjugate in $\text{MCG}(\Sigma_2)$:

$$\rho(w_B) = \rho(s_1)^{-1}\rho(w_A)\rho(s_1).$$

- (3) **Topological Entropy Equality:** Topological entropy is invariant under mapping class conjugacy, yielding exact equality:

$$h_{\text{top}}(\rho(w_A)) = h_{\text{top}}(\rho(w_B)).$$

- (4) **3-Manifold Homeomorphism:** The mapping tori associated with $\rho(w_A)$ and $\rho(w_B)$ are homeomorphic as closed 3-manifolds:

$$M_{\rho(w_A)} \cong M_{\rho(w_B)}.$$

Proof. Part (1) follows from the non-commutativity of the generators s_1, s_2 .

For Part (2), expanding the conjugate word yields $s_1^{-1}w_A s_1 = s_1^{-1}(s_1 s_2)s_1 = s_2 s_1 = w_B$. Applying the group homomorphism property $\rho(gh) = \rho(g)\rho(h)$ established in Section 4 gives $\rho(w_B) = \rho(s_1)^{-1}\rho(w_A)\rho(s_1)$.

For Part (3), topological entropy is a dynamical invariant under topological conjugacy. For any homeomorphism $f : \Sigma_2 \rightarrow \Sigma_2$ and conjugating homeomorphism $h : \Sigma_2 \rightarrow \Sigma_2$, $h_{\text{top}}(h^{-1} \circ f \circ h) = h_{\text{top}}(f)$, whence $h_{\text{top}}(\rho(w_A)) = h_{\text{top}}(\rho(w_B))$.

For Part (4), choose representative homeomorphisms ϕ_A, ϕ_B, ϕ_1 for $\rho(w_A), \rho(w_B), \rho(s_1)$ satisfying $\phi_B = \phi_1^{-1} \circ \phi_A \circ \phi_1$, which implies $\phi_1^{-1} \circ \phi_A = \phi_B \circ \phi_1^{-1}$. We define a continuous map $H : \Sigma_2 \times [0, 1] \rightarrow \Sigma_2 \times [0, 1]$ by $H(x, t) = (\phi_1^{-1}(x), t)$. At the quotient boundaries,

$$H(x, 0) = (\phi_1^{-1}(x), 0) \quad \text{and} \quad H(\phi_A(x), 1) = (\phi_1^{-1}(\phi_A(x)), 1) = (\phi_B(\phi_1^{-1}(x)), 1).$$

This respects the quotient boundary identification $(x, 0) \sim (\phi_A(x), 1)$ in $M_{\rho(w_A)}$ and maps it directly to $(y, 0) \sim (\phi_B(y), 1)$ in $M_{\rho(w_B)}$ under $y = \phi_1^{-1}(x)$. Hence H descends to a well-defined global homeomorphism $\bar{H} : M_{\rho(w_A)} \xrightarrow{\cong} M_{\rho(w_B)}$, establishing $M_{\rho(w_A)} \cong M_{\rho(w_B)}$. $\square$

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